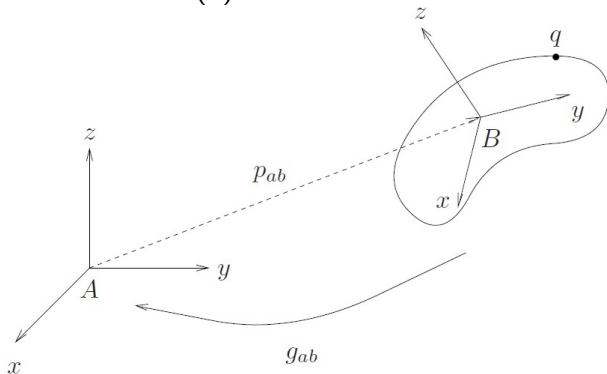


§3 Rigid Motion in $\mathbb{R}^3$

$p_{AB} \in \mathbb{R}^3$ and $R \in SO(3)$

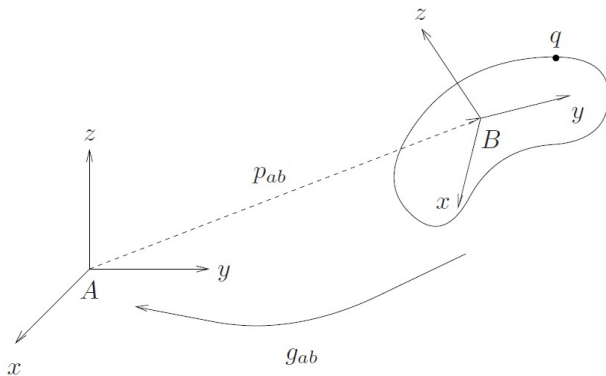


$$SE(3) = \{(p, R) : p \in \mathbb{R}^3, R \in SO(3)\} = \mathbb{R}^3 \times SO(3) \quad (2.21)$$

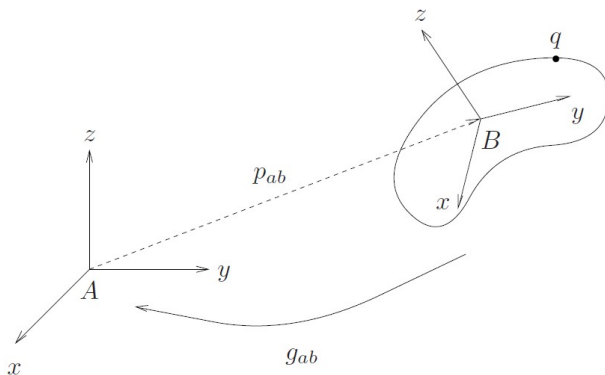
§3 Rigid Motion in $\mathbb{R}^3$

$(p, R) \in SE(3)$ serves as both

- ▶ a specification of configuration of a rigid body and
- ▶ a transformation coordinates from one frame to another.



§3 Rigid Motion in $\mathbb{R}^3$



Transformation: Let $q_a, q_b \in \mathbb{R}^3$ be the coordinates of q in relative to frames A and B , respectively.

$$q_a = p_{ab} + R_{ab}q_b \quad (2.22)$$

where $g_{ab} = (p_{ab}, R_{ab}) \in SE(3)$ specifies configuration of frame B .

§3 Rigid Motion in $\mathbb{R}^3$

While $g_{ab} \in SE(3)$ specifies the configuration of frame B , we also use g to indicate the action of a rigid transformation, therefore, from eq. (2.22):

$$q_a = p_{ab} + R_{ab}q_b = g_{ab}(q_b)$$

§3.1 Homogeneous representation

The *homogeneous coordinates* of point q , in $\mathbb{R}^4$, is defined as

$$\bar{q} = \begin{bmatrix} q_1 \\ q_2 \\ q_3 \\ 1 \end{bmatrix}$$

§3.1 Homogeneous representation

On the other hand, a vector v with components $[v_1 \ v_2 \ v_3]^T \in \mathbb{R}^3$ is expressed in homogeneous coordinates in $\mathbb{R}^4$ as

$$\bar{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ 0 \end{bmatrix}$$

§3.1 Rules of syntax for homogeneous representation

1. Sums and differences of vectors are vectors.
2. The sum of a vector and a point is a point.
3. The difference between two points is a vector.
4. The sum of two points is meaningless.

§3.1 Homogeneous representation

The homogeneous representation allows us to write the *affine transformation* $q_a = g_{ab}(q_b)$ in linear form:

$$\bar{q}_a = \begin{bmatrix} q_a \\ 1 \end{bmatrix} = \begin{bmatrix} R_{ab} & p_{ab} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} q_b \\ 1 \end{bmatrix} = \bar{g}_{ab} \bar{q}_b$$

where the 0 symbol in the matrix $\bar{g}_{ab}$ indicate a 1×3 zero matrix, while the 1 is a scalar 1.

$\bar{g}_{ab} \in \mathbb{R}^{4 \times 4}$ is called the homogeneous representation of $g_{ab} \in SE(3)$.

Composition rule for rigid body transformations

Let $g_{bc} \in SE(3)$ be the configuration of frame C relative to frame B and $g_{ab} \in SE(3)$ be the configuration of frame B relative to frame A . Then the configuration of frame C relative to frame A can be extracted from

$$\bar{g}_{ac} = \bar{g}_{ab}\bar{g}_{bc} = \begin{bmatrix} R_{ab}R_{bc} & R_{ab}p_{bc} + p_{ab} \\ 0 & 1 \end{bmatrix} \quad (2.24)$$

provided that

- ▶ R_{ab} is expressed in components in frame A and R_{bc} is expressed in components in frame B .
- ▶ p_{ab} is expressed in components in frame A and p_{bc} is expressed in components in frame B .

Eq. (2.24) is the *composition rule* for rigid body transformations

The set of rigid transformations is a group

See p. 37

For point 4: MATLAB demo

Rigid transformation in homogeneous form applied to a vector

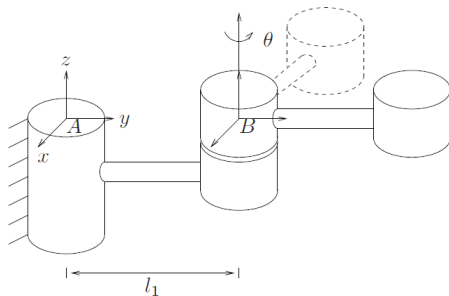
Rigid transformation of vector $v = s - r$, s and r points

$$\bar{g}_* \bar{v} = \bar{g}(\bar{s}) - \bar{g}(\bar{r}) = \begin{bmatrix} R & p \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ 0 \end{bmatrix}$$

MLS now simplify the notation:

write gq and gv instead of $\bar{g}\bar{q}$ and $\bar{g}_*\bar{v}$

Example 2.1



Rotation θ about line in z direction through $(0, l_1, 0) \in \mathbb{R}^3$.

§3.2 Exponential coordinates for rigid motion and twists

Note introductory paragraph

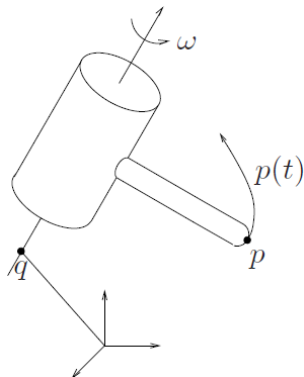
Axis of rotation $\omega \in \mathbb{R}^3, \|\omega\| = 1$

$q \in \mathbb{R}^3$ on axis

As before, assume rotation with unit angular velocity.

Eq. (2.25)

$$\dot{p}(t) = \hat{\omega}(p(t) - q)$$



Revolute joint

§3.2 Exponential coordinates for rigid motion and twists

$$\dot{p}(t) = \hat{\omega}(p(t) - q) \quad (2.25)$$

We now define (and note new additional notation)

$$\hat{\xi} = \begin{bmatrix} \hat{\omega} & v \\ 0 & 0 \end{bmatrix} \quad (2.26)$$

where $v = -\hat{\omega}q$. This allows us to write eqn. (2.25) in homogeneous form:

$$\begin{bmatrix} \dot{p} \\ 0 \end{bmatrix} = \begin{bmatrix} \hat{\omega} & -\hat{\omega}q \\ 0 & 0 \end{bmatrix} \begin{bmatrix} p \\ 1 \end{bmatrix} = \hat{\xi} \begin{bmatrix} p \\ 1 \end{bmatrix} \Rightarrow \dot{\bar{p}} = \hat{\xi} \bar{p}$$

§3.2 Exponential coordinates for rigid motion and twists

$$\dot{\bar{p}} = \hat{\xi} \bar{p}$$

Solution to the ODE:

$$\bar{p}(t) = e^{\hat{\xi}t} \bar{p}(0)$$

where

$$e^{\hat{\xi}t} = I + \hat{\xi}t + \frac{(\hat{\xi}t)^2}{2!} + \frac{(\hat{\xi}t)^3}{3!} + \dots$$

$e^{\hat{\xi}t}$ is a mapping from the initial location of a point to its location after rotating t radians.

§3.2 Exponential coordinates for rigid motion and twists

Translation: velocity of point attached to a prismatic joint moving with unit velocity:

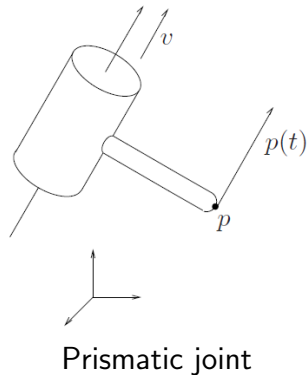
$$\dot{p}(t) = v \quad (2.27)$$

As before, this can be written as

$$\begin{bmatrix} \dot{p} \\ 0 \end{bmatrix} = \hat{\xi} \begin{bmatrix} p \\ 1 \end{bmatrix} \Rightarrow \dot{\bar{p}} = \hat{\xi} \bar{p}$$

where

$$\hat{\xi} = \begin{bmatrix} 0_{3 \times 3} & v \\ 0_{1 \times 3} & 0 \end{bmatrix} \quad (2.28)$$



§3.2 Exponential coordinates for rigid motion and twists

$$\dot{\bar{p}} = \hat{\xi} \bar{p}$$

As before, the solution to this ODE:

$$\bar{p}(t) = e^{\hat{\xi}t} \bar{p}(0)$$

where t is the total amount of the translation.

MLS define

$$se(3) := \{(v, \hat{\omega}) : v \in \mathbb{R}^3, \hat{\omega} \in so(3)\} \quad (2.29)$$

Homogeneous coordinates: $\hat{\xi} \in se(3)$ is written as

$$\hat{\xi} = \begin{bmatrix} \hat{\omega} & v \\ 0 & 0 \end{bmatrix} \in \mathbb{R}^{4 \times 4}$$

An element of $se(3)$ is called a *twist*.

§3.2 Exponential coordinates for rigid motion and twists

Vee and wedge operators:

$$\begin{bmatrix} \hat{\omega} & \mathbf{v} \\ 0 & 0 \end{bmatrix}^{\vee} = \begin{bmatrix} \mathbf{v} \\ \omega \end{bmatrix} \quad (2.30)$$

$$\begin{bmatrix} \mathbf{v} \\ \omega \end{bmatrix}^{\wedge} = \begin{bmatrix} \hat{\omega} & \mathbf{v} \\ 0 & 0 \end{bmatrix} \quad (2.31)$$

RHS of eqn. (2.30) is $\xi \in \mathbb{R}^6$, the *twist coordinates* of the twist $\hat{\xi} \in se(3)$.

§3.2 Exponential coordinates for rigid motion and twists

Proposition 2.8 (p. 41): given $\hat{\xi} \in se(3)$ and $\theta \in \mathbb{R}$, $e^{\hat{\xi}\theta} \in SE(3)$.

$$\hat{\xi} = \begin{bmatrix} \hat{\omega} & \mathbf{v} \\ 0 & 0 \end{bmatrix}$$

Case 1: $\omega = 0$; this leads to [eq. \(2.32\)](#), which confirms that the proposition is true for case 1 (pure translation). Work through this in detail.

Case 2: $\omega \neq 0$. Assume $\|\omega\| = 1$, by appropriate scaling of θ , if necessary.

Define a rigid transformation

$$g = \begin{bmatrix} I & \hat{\omega} v \\ 0 & 1 \end{bmatrix} \quad (2.33)$$

We can define & calculate

$$\begin{aligned} \hat{\xi}' &= g^{-1} \hat{\xi} g \\ &= \begin{bmatrix} I & -\hat{\omega} v \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \hat{\omega} & v \\ 0 & 0 \end{bmatrix} \begin{bmatrix} I & \hat{\omega} v \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} \hat{\omega} & \omega \omega^T v \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} \hat{\omega} & h \omega \\ 0 & 0 \end{bmatrix} \end{aligned} \quad (2.34)$$

where we used lemma 2.3 on p. 28, with $||\omega|| = 1$, and where $h := \omega^T v$.

We now use the identity:

$$e^{\hat{\xi}\theta} = g e^{\hat{\xi}'\theta} g^{-1} \quad (2.35)$$

because it is much easier to calculate $e^{\hat{\xi}'\theta}$ than $e^{\hat{\xi}\theta}$.

Realizing that $\hat{\omega}\omega = 0$,

$$(\hat{\xi}')^2 = \begin{bmatrix} \hat{\omega} & h\omega \\ 0 & 0 \end{bmatrix}^2 = \begin{bmatrix} \hat{\omega}^2 & 0 \\ 0 & 0 \end{bmatrix}$$

$$(\hat{\xi}')^3 = \begin{bmatrix} \hat{\omega} & h\omega \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \hat{\omega}^2 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} \hat{\omega}^3 & 0 \\ 0 & 0 \end{bmatrix} \dots \text{etc.}$$

$$e^{\hat{\xi}'\theta} = I + \hat{\xi}'\theta + \frac{(\hat{\xi}'\theta)^2}{2!} + \frac{(\hat{\xi}'\theta)^3}{3!} \dots$$

With

$$\hat{\xi}' = \begin{bmatrix} \hat{\omega} & h\omega \\ 0 & 0 \end{bmatrix}$$

$$(\hat{\xi}')^2 = \begin{bmatrix} \hat{\omega}^2 & 0 \\ 0 & 0 \end{bmatrix}, \quad (\hat{\xi}')^3 = \begin{bmatrix} \hat{\omega}^3 & 0 \\ 0 & 0 \end{bmatrix} \dots \text{etc.}$$

it follows that

$$e^{\hat{\xi}'\theta} = \begin{bmatrix} e^{\hat{\omega}\theta} & h\omega\theta \\ 0 & 1 \end{bmatrix}$$

$$e^{\hat{\xi}'\theta} = \begin{bmatrix} e^{\hat{\omega}\theta} & h\omega\theta \\ 0 & 1 \end{bmatrix}$$

Now using identity (2.35) gives

$$\begin{aligned} e^{\hat{\xi}\theta} &= g e^{\hat{\xi}'\theta} g^{-1} \\ &= \begin{bmatrix} I & \hat{\omega}v \\ 0 & 1 \end{bmatrix} \begin{bmatrix} e^{\hat{\omega}\theta} & h\omega\theta \\ 0 & 1 \end{bmatrix} \begin{bmatrix} I & -\hat{\omega}v \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} e^{\hat{\omega}\theta} & (I - e^{\hat{\omega}\theta})\hat{\omega}v + \omega\omega^T v\theta \\ 0 & 1 \end{bmatrix} \in SE(3) \quad \omega \neq 0 \quad (2.36) \end{aligned}$$

Transformation $g = e^{\hat{\xi}\theta}$

$$e^{\hat{\xi}\theta} = \begin{bmatrix} e^{\hat{\omega}\theta} & (I - e^{\hat{\omega}\theta})\hat{\omega}v + \omega\omega^T v\theta \\ 0 & 1 \end{bmatrix} \quad (2.36)$$

- ▶ Eqn. (2.36) provides us with a easy way to calculate $e^{\hat{\xi}\theta}$, since $e^{\hat{\omega}\theta}$ can be very easily calculated by eqn. (2.14) p. 28 (Rodrigues' formula)
- ▶ $p(\theta) = e^{\hat{\xi}\theta}p(0)$ mapping from initial to final coordinates of p in same frame
- ▶ $g_{ab}(\theta) = e^{\hat{\xi}\theta}g_{ab}(0)$ (eqn. (2.37)) mapping from initial to final configuration of body in frame A

§3.2 Exponential coordinates for rigid motion and twists

Proposition 2.9: Given $g \in SE(3)$, there exists $\hat{\xi} \in se(3)$ and $\theta \in \mathbb{R}$ such that $g = e^{\hat{\xi}\theta}$.

Consider $g = (R, p)$ with $R \in SO(3)$, $p \in \mathbb{R}^3$

Trivial case: $(R, p) = (I, 0) \Leftrightarrow \theta = 0$, arbitrary $\hat{\xi}$

Case 1: $R = I$ (that is, no rotational motion)

$$\hat{\xi} = \begin{bmatrix} 0 & \frac{p}{\|p\|} \\ 0 & 0 \end{bmatrix} \quad \theta = \|p\|$$

Similar to the derivation of eqn. (2.32):

$$e^{\hat{\xi}\theta} = I + \hat{\xi}\theta = (I, p) = g$$

§3.2 Exponential coordinates for rigid motion and twists

Proposition 2.9: Case 2: $R \neq I$

To find $\xi = (v, \omega)$, we set $e^{\hat{\xi}\theta} = g$ and solve for (v, ω) .

Using eq. (2.36):

$$e^{\hat{\xi}\theta} = \begin{bmatrix} e^{\hat{\omega}\theta} & (I - e^{\hat{\omega}\theta})\hat{\omega}v + \omega\omega^T v\theta \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} R & p \\ 0 & 1 \end{bmatrix}$$

ω and θ is solved by setting $e^{\hat{\omega}\theta} = R$ and then using proposition 2.5 p. 29. This leaves

$$(I - e^{\hat{\omega}\theta})\hat{\omega}v + \omega\omega^T v\theta = p \quad (2.38)$$

to solve v .

§3.2 Exponential coordinates for rigid motion and twists

Proposition 2.9: Case 2: $R \neq I$

$$(I - e^{\hat{\omega}\theta})\hat{\omega}v + \omega\omega^T v\theta = p \quad (2.38)$$

Eqn. (2.38) can be written as

$$\left[(I - e^{\hat{\omega}\theta})\hat{\omega} + \omega\omega^T\theta \right] v = Av = p$$

where

$$A = (I - e^{\hat{\omega}\theta})\hat{\omega} + \omega\omega^T\theta$$

It can be shown that for $R \neq I$ and $\theta \neq 0$, $|A| \neq 0$, hence v can be calculated for a given p .

$$e^{\hat{\xi}\theta} = \begin{bmatrix} I & \nu\theta \\ 0 & 1 \end{bmatrix} \qquad \omega = 0 \qquad (2.32)$$

Proof of identity (2.35)

We defined $\hat{\xi}' = g^{-1}\hat{\xi}g$. Therefore $\hat{\xi} = g\hat{\xi}'g^{-1}$.

$$\begin{aligned}e^{\hat{\xi}\theta} &= e^{g\hat{\xi}'g^{-1}\theta} = e^{g\hat{\xi}'\theta g^{-1}} \\&= I + g\hat{\xi}'\theta g^{-1} + \frac{(g\hat{\xi}'\theta g^{-1})^2}{2!} + \frac{(g\hat{\xi}'\theta g^{-1})^3}{3!} \dots \\&= I + g\hat{\xi}'\theta g^{-1} + g\frac{(\hat{\xi}'\theta)^2}{2!}g^{-1} + g\frac{(\hat{\xi}'\theta)^3}{3!}g^{-1} \dots \\&= g\left(I + \hat{\xi}'\theta + \frac{(\hat{\xi}'\theta)^2}{2!} + \frac{(\hat{\xi}'\theta)^3}{3!} \dots\right)g^{-1} \\&= ge^{\hat{\xi}'\theta}g^{-1}\end{aligned}$$